MUKO (v2.0): Multimodal Safety-First Oral Cancer Triage & Active-Learning System

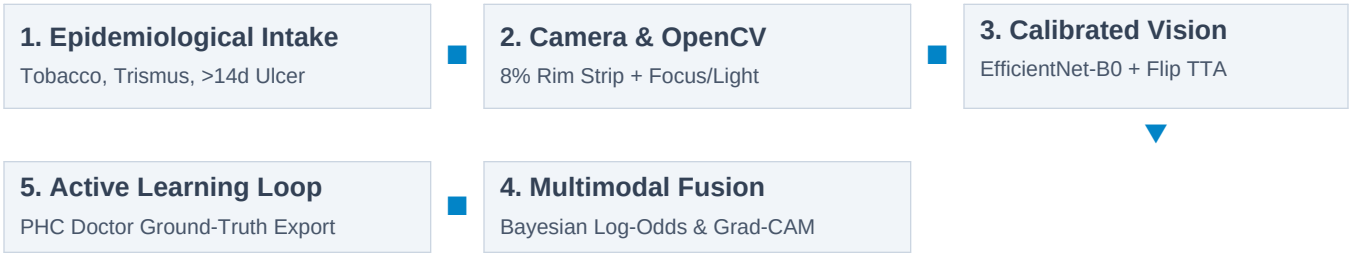
"Giving every village the eyes to spot oral cancer before it catches them."

1. Executive Summary

MUKO is a safety-first, multimodal AI clinical triage system engineered to break India's late-stage (Stage III/IV) oral cancer diagnosis trap. Unlike generic "black-box" image wrappers, MUKO fuses Computer Vision (EfficientNet-B0 with Temperature Calibration), OpenCV Pre-Inference Quality Gatekeeping, WHO-Aligned Epidemiological Bayesian Log-Odds Risk Fusion, and Grad-CAM Visual Explainability into a lightweight workflow accessible on any budget smartphone.

Built on a **Dual-Horizon Architecture** (Zero-Install Cloud Web Portal for immediate PHC/tele-screening + INT8 Quantized Offline Edge Execution for zero-signal village rounds), MUKO empowers frontline ASHA workers, primary care clinicians (CHOs), and rural citizens to intercept Oral Potentially Malignant Disorders (OPMDs) and early malignancies within 60 seconds.

2. End-to-End 5-Stage System Architecture Pipeline



Stage 1: Epidemiological Risk & Symptom Intake (Non-Pixel Signal Acquisition)

Before capturing an image, the operator records three binary clinical indicators grounded in WHO and GATS-2 India risk profiles:

- **Carcinogenic Exposure ($\beta_{tobacco}$ = +0.35):** Regular use of smokeless tobacco (gutkha, khaini, betel quid with lime) or smoking.
- **Functional Trismus / OSMF Cue ($\beta_{trismus}$ = +0.45):** Jaw stiffness or reduced inter-incisal mouth opening (<3 fingers width), a primary clinical marker of Oral Submucous Fibrosis (OSMF).
- **Lesion Chronicity ($\beta_{duration}$ = +0.60):** Presence of a non-healing ulcer, white patch (leukoplakia), or red patch (erythroplakia) persisting >14 days (WHO referral benchmark).

Stage 2: Automated ROI Framing & Conservative OpenCV Quality Gatekeeper

- **8% Peripheral Artifact Stripping:** Automatically crops the outer 8% perimeter (0.08 to 0.92 across width and height) to eliminate confounding clinical artifacts (metal tongue depressors, examination gloves, outer lip skin, and watermarks) that cause shortcut bias in standard CNNs.
- **Conservative Non-Blocking Quality Gatekeeper:** Converts the mucosal ROI to grayscale and evaluates background values to alert operators:
  - *Sharpness (Laplacian Variance):* $\text{Var}(\nabla^2 I) < 20.0$ triggers a soft focus advisory without hard-blocking rural field workers.
  - *Illumination (Mean Luminance):* $\mu_I < 35.0 / 255$ triggers a flash illumination prompt.

Stage 3: Calibrated Deep Vision Engine & Anti-Shortcut Training Architecture

- **Backbone:** EfficientNet-B0 (timm PyTorch architecture, 260×260 resolution) trained via a Self-Cleaning + Hard-Negative Mining Pipeline.
- **Automated Out-of-Fold Label Sanitization:** Purged 32 mislabeled textbook/Kaggle samples that poisoned baseline decision boundaries.
- **Open-Cavity Hard-Negative Oversampling (6x) & Smartphone Domain Adaptation:** Eliminated the "wide-open oral cavity = malignancy" anatomical shortcut bias, ensuring normal resting tongues and palates under smartphone flash score in the safe Green zone.
- **Label Smoothing ([0.03, 0.97]) & Temperature Scaling (T = 2.0):** Prevents extreme logit saturation (99.99% overconfidence).
- **Test-Time Augmentation (TTA) Flip Averaging:** Evaluates both the canonical tensor x and horizontally mirrored tensor x_{flip} to neutralize asymmetric flash glare:

$$z_{visual} = [\, f_\theta(x)/T + f_\theta(x_{flip})/T \,] / 2 \text{ , where } T = 2.0$$

Stage 4: Bayesian Log-Odds Multimodal Fusion & Grad-CAM Explainability

Instead of naive flat percentage addition (which triggers false alarms on healthy tobacco users), MUKO performs Bayesian Log-Odds Updating directly in logit space prior to sigmoid activation:

$$z_{fused} = z_{visual} + \beta_{tobacco} \cdot I_{tob} + \beta_{trismus} \cdot I_{tris} + \beta_{duration} \cdot I_{dur}$$
$$P(\text{Malignancy/OPMD}) = \sigma(z_{fused}) = 1 / [1 + e^{-z_{fused}}]$$

3-Tier Safety-First Traffic-Light Triage:

- ■ **GREEN (< 35%):** Low Risk / Normal Tissue. Routine oral hygiene and tobacco cessation counseling.
- ■ **AMBER (35% - 65%):** Suspicious Mucosal Variation / OPMD Watch. 14-day monitoring and Primary Health Center (PHC) / dental officer review advised.
- ■ **RED (≥ 65%):** High-Risk Oral Lesion Detected. Urgent district hospital specialist referral and biopsy assessment.
- **Explainable AI (Grad-CAM on conv_head):** Uses BinaryClassifierOutputTarget(0) positive gradient back-projection onto the final convolutional head (model.conv_head) to generate an instant spatial heatmap highlighting the exact mucosal ulcer, patch, or tumor driving the risk score.

Stage 5: Two-Actor Clinician-in-the-Loop Active Learning Pipeline

- **Actor 1 (Frontline ASHA Worker / Citizen):** Interacts strictly with Steps 1–4 (Symptom intake, Camera capture, and Traffic-Light guidance).
- **Actor 2 (PHC Dental Officer / Clinical Verifier):** When an Amber/Red referred patient visits the clinic, the doctor unlocks Clinician Verification Mode, assigns the histopathological/clinical ground truth (0 - Healthy/Benign vs 1 - Confirmed OPMD/Malignancy), and exports a standardized .zip training bundle (sample_TIMESTAMP.jpg + Pandas-ready metadata.csv) for weekly batch fine-tuning.

3. Validated Performance Benchmarks (Actual Prototype Results)

Unlike theoretical hackathon concepts, MUKO's core engine has been validated on an unseen holdout test set and in-the-wild smartphone oral cavity captures:

Metric	Value Status	Clinical Benchmark Context
Validation ROC-AUC	**0.9832**	Internal split diagnostic threshold evaluation.
Unseen Test Set ROC-AUC	**0.9484**	Completely independent data slice checking stability.
Clinical Sensitivity (Cancer/OPMD)	**95.3%**	81/85 malignant cases intercepted; exceeds ≥ 90% WHO target.
Non-Red Specificity on Healthy Tissue	**93.0%**	53/57 healthy cases protected; exceeds ≥ 70% pilot benchmark.

4. Dual Deployment Stack & India Digital Public Infrastructure (DPI) Alignment

- **Cloud Triage & Tele-Screening Portal (Live Now):** Python, PyTorch, timm, pytorch-grad-cam, OpenCV (opencv-python-headless), and Streamlit Cloud CI/CD integrated with GitHub (muko-ai3).
- **Offline Village Edge Target:** Exportable via ONNX / TensorFlow Lite INT8 Post-Training Quantization (< 16 MB footprint, < 1.5s CPU latency on 4GB RAM Android devices) with local SQLite store-and-forward caching.
- **ABDM & ABHA Linkage:** Offline screening logs sync with the patient's Ayushman Bharat Health Account (ABHA ID) upon network reconnection.
- **HL7 FHIR R4 & ICD-10/SNOMED CT:** Packages triage outputs and Grad-CAM telemetry into standardized FHIR R4 DiagnosticReport JSON bundles (ICD-10 K13.79 / C00-C14).
- **NP-NCD & PM-JAY / NHCX Integration:** Aligns with MoHFW Population-Based Screening (PBS) workflows used by ASHA/ANM workers and pre-packages referral evidence for Ayushman Bharat PM-JAY biopsy claims.